

Alireza Irandoost

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EDUCATION

Sharif University of Technology, Tehran, Iran M.Sc. in Micro and Nanoelectronic Devices – GPA: 18.06/20 (4/4)	October 2021 – January 2024
University of Tehran, Tehran, Iran B.Sc. in Electrical Engineering (Electronics) – GPA: 17.56/20 (3.78/4)	September 2016 – July 2021
Farabi High School, Qazvin, Iran Diploma in Mathematics and Physics – GPA: 19.77/20	Sep. 2012 – Jul. 2016

RESEARCH INTERESTS

◇ Biosensors	◇ Micro/nanofabrication	◇ Electronic Devices	◇ High-resolution Microscopy
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RESEARCH & WORK EXPERIENCES

Research Assistant (Remote) ◇ <i>Bio-Inspired Sensors and Optoelectronics Laboratory</i> – Supervisor: Prof. Hooman Mohseni University, USA	July 2024 – January 2025 Northwestern
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- Worked on analytical modeling of heterojunction single-photon detectors and simulation of optical communication modes to identify the optimal optical channels in the lossy media and to construct the desired beam shape.

Research Assistant ◇ <i>Superconductor Electronics Research Lab (SERL)</i> – Supervisor: Prof. Mehdi Fardmanesh Technology, Iran	October 2021 – Present Sharif University of
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- Hands-on experience in the cleanroom environment and proficiency in various fabrication processes such as photolithography, thin film deposition, wet etching, electroplating, layout design, and blade dicing.
- Process development for fabrication of Silicon-based and flexible neural electrodes, Mach-Zehnder Modulator, and Micro Electrode Array (MEA).
- Utilizing characterization techniques such as fourier-transform infrared spectroscopy (FTIR), electrochemical impedance spectroscopy (EIS), and four-point probe.
- Performing scientific experiments for undergraduate students in the lab and leading two of them to do their B.Sc. projects.

Researcher ◇ <i>Institute for Research and Fundamental Science (IPM)</i> – Supervisor: Prof. Mehdi Fardmanesh	December 2022 – January 2024 Tehran, Iran
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- Low-noise measurements for neural signal recording and signal processing of the recorded data for spike detection and local field potential (LFP) signals.

Hardware Engineer ◇ <i>Persian Gazelle Solar Car IV (PGSC)</i> – Supervisor: Prof. Karen Abrinia	July 2018 – December 2019 University of Tehran, Iran
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- Implemented STM32 ARM-based electronic circuit for controlling brushless DC motors and developed CAN bus between electronic boards of the car to communicate with each other.

Internship ◇ <i>Irankhodro Powertrain Co.(IPCO)</i> – Research & Design Automotive Company	Summer 2019 Tehran, Iran
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- Designed and built a wireless data logger for engine control unit and developed a user interface software for monitoring data with C#.

NOTABLE PROJECTS

- ◇ **M.Sc. Thesis – Design and Fabrication of Microelectrode-based Neural Probe**, Supervisor: Prof. Fardmanesh. Process development for fabrication of flexible and silicon-based neural probes for deep brain implantation, characterization of them using electrochemical impedance spectroscopy, and recording the in-vivo local field potential signals in the 0.5 to 7 Hz frequency band from the brain of a zebra finch bird in response to sound stimulation.
- ◇ Design and Implementation of **Kaufman Ion Source for Ion Beam Etching**, ongoing, Supervisor: Prof. Fardmanesh. Developed an electron-beam using thermionic emission from a hot filament to generate plasma with impact ionization and accelerate ions for ion-beam etching.
- ◇ Participated in **Fabrication of Micro Electrode Array (MEA)**, Supervisor: Prof. Fardmanesh. Performing thin film deposition of Au/Ti using PVD, photolithography, and HF and aqua regia etchants to make 64 microelectrode arrays for in-vitro neural cells recording.

- ◇ Participated in **Fabrication of Mach-Zehnder Modulator**, Supervisor: Prof. Fardmanesh. Using LiNbO₃ substrate, deposition of 200 nm Ti, photolithography, HF etching, Ti indiffusion at 1050 °C, and fabrication of
- ◇ B.Sc. Project – Design and Simulation of **Electrical Driver for Brushless DC Machines**, Supervisor: Prof. Imaneini. Designed a 120 W electronic driver using a new method for sensorless control of brushless DC machines.
- ◇ Simulation of **Wide-field Fluorescence Microscopy**, with a resolution of 180 nm for imaging stained cells using Zemax OpticStudio.
- ◇ Designed and built an optical setup for **Surface Plasmon Resonance (SPR)** biosensor using Kretschmann configuration, 655 nm laser diode, linear polarizer, beam profiler, and optical lens to study biomolecular interactions.
- ◇ Simulation of **Surface Acoustic Wave**, Supervisor: Prof. Abdollahad. Using LiNbO₃ substrate as a piezoelectric material and gold electrodes on top of it to excite the acoustical field for heating cancer cells in COMSOL.

Course Projects

- ◇ Simulation of **Optical Ring Resonator, Mach-Zehnder Modulator, and Fabry-Perot Laser** in Ansys Lumerical, Optoelectronics, Instructor: Prof. Kavevash
- ◇ Design and Simulation of **Avalanche Photodetector, PN Diode, and MESFET** using Silvaco TCAD, Advanced Solid-state Devices, Instructor: Prof. Rashidan
- ◇ Design and simulation of **Differential Amplifier**, with specifications of BW = 145.7 MHz, gain = 60.33 dB, phase margin = 65°, and total power = 0.761 mW in 180 nm CMOS technology using **Cadence Virtuoso**, Electronics III lab, Instructor: Prof. Sheikhaei
- ◇ Process design and simulation for **Double Diffusion Repeatability of P-Type Wafer** for fabrication of PNP BJT transistor in Silvaco TCAD, Semiconductor Technology, Instructor: Prof. Rashidan
- ◇ Implemented **Single Cycle, Multi Cycle, and Pipeline MIPS Processor** and two layer **Neural Network** using Verilog, Computer Architecture, Instructor: Prof. Safari
- ◇ Developed **Multi-agent Pac-Man Game**, Artificial Intelligence, Instructor: Prof. MoradiSabzevar
Using AI search-based algorithms such as breadth-first search, iterative deepening search, A* search in Python.
- ◇ Implemented **Naïve Bayes Base Classifier** for classification of Hafez and Saadi poems in Python, Artificial Intelligence, Instructor: Prof. MoradiSabzevar
- ◇ Designed discrete digital filters for **ECG signals** for finding heart beating rate and **Image Signal Processing** using Kernel matrix in MATLAB, Digital Signal Processing, Instructor: Prof. Badieirostam
- ◇ Design and simulation of **Buck-boost, Boost, and Flyback Converters** in Simulink and Pspice, Industrial Electronics, Instructor: Prof. Imaneini
- ◇ Implemented **Multi Purpose Function Generator Using FPGA**, Logic Circuit Lab, Instructor: Prof. Navabi
Generated waveforms have different frequencies, amplitudes, and shapes such as sine, square, rhomboid, and sawtooth.

PUBLICATION

- ◇ **A. Irandoost, R. Mohajeri, A. Bahramani, A. GhaziZadeh, and M. Fardmanesh, A novel cost-effective fabrication of a flexible neural probe for brain signal recording**, <https://doi.org/10.48550/arXiv.2509.09213>
- ◇ **A. Irandoost, R. Mohajeri, A. Bahramani, A. GhaziZadeh, and M. Fardmanesh, Fabrication of a Low-cost Multi-electrode Neural Probe for Brain Signal Recording**. DOI: 10.1109/ICBME61513.2023.10488585

HONORS & AWARDS

- ◇ Awarded Murphy fellowship for Ph.D. program in Electrical Engineering from Northwestern University, 2024
- ◇ Admitted to PhD program of Sharif University of Technology without entrance exam as an exceptional talented student, 2024
- ◇ Best MSc Poster Award, Sharif University of Technology, 2023
- ◇ Best Paper and Presentation Award, ICBME and Sharif University of Technology, 2023
- ◇ Received scholarship as a talented student from the supporter foundation of University of Tehran, 2020
- ◇ Won Faculty of Engineering (FOE) Award, University of Tehran, 2017

SELECTED COURSES

Graduate

Optoelectronics, 17.7/20
 Applied Quantum Mechanics, 19.1/20
 Advanced Solid-state Devices, 17/20
 Superconductivity Principles, 18.5/20
 Theory and Technology of Electronic Devices Fabrication, 19.2/20
 Advanced Solid-state Physics, 16.5/20
 Medical Ultrasound, 19.2/20

Undergraduate

Electronic Physics, 19/20
 Modern Physics, 20/20
 Electromagnetism, 17.4/20
 Artificial Intelligence, 17.5/20
 Electronics I, II, III
 Filter and Circuit Synthesis, 20/20
 Digital Electronics, 16.5/20

TEACHING EXPERIENCE

Teaching Assistance

Spring 2023, Medical Ultrasound, Prof. Kavehvasht
Fall 2022, Principle of Solid-state Physics, Prof. Rashidian
Fall 2022, Advanced Solid-state Devices, Prof. Rashidian

Spring 2022/23, Biosensor Lab, Prof. Fardmanesh
Fall 2022, Semiconductor Technology, Prof. Rashidian
Fall 2020, Electronic Physics, Prof. Mohajerzadeh

TECHNICAL SKILLS

Programming Software

C/C++, Python, Verilog
MATLAB, Advanced Design System (ADS), COMSOL, AutoCAD, NI LabVIEW, Ansys
Lumerical (FDTD), Silvaco TCAD, HSPICE, Altium Designer (PCB design), Altera Quartus, ModelSim

LANGUAGES

Persian (Native)

English (TOEFL 92/120, R:24, L:29, S:20, W:19)

Azerbaijani (Fluent)

REFERENCES

Prof. Mehdi Fardmanesh, Professor at School of Electrical Engineering, Sharif University of Technology, Tehran, Iran. E-mail: fardmanesh@sharif.edu

Prof. Bizhan Rashidian, Distinguished Professor at School of Electrical Engineering, Sharif University of Technology, Tehran, Iran. E-mail: rashidia@sharif.edu

Prof. Shamsodin Mohajerzadeh, Full Professor at School of Electrical and Computer Engineering, University of Tehran, Tehran, Iran. E-mail: mohajer@ut.ac.ir